MARCELO CARVALHO FILHO

SITE RELIABILITY ENGINEER | KUBERNETES · AWS · TERRAFORM · GITOPS

Palm Bay, Florida, USA | Relocating to Spain, summer 2027 (employer sponsorship required)
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CAREER HIGHLIGHTS

Reliability: Escalation lead for 24/7 incident response across a 1,200+ server multi-tenant fleet; root-cause analysis and vulnerability remediation helped cut monthly outages from about 10 to 3–4 and MTTR by 25%.

Automation: Ansible and Jenkins automation modernized 200+ servers (60% less manual work) and cut new environment deployment time from 18.8 to 5.6 days. Led 100+ production website migrations with controlled 2-hour cutover windows

Platform Engineering: Run a 4-node K3s cluster with ArgoCD GitOps and a 3-node OpenStack private cloud in a self-hosted lab; Terraform and GitHub Actions deploy my production portfolio site on AWS.

Credentials: 4 years of full-time infrastructure experience, built on earlier IT consulting. AWS Solutions Architect – Associate, HashiCorp Terraform Associate, CompTIA Security+ and Cloud+; Portuguese, English, Spanish (C1), and Italian (B2).

AREAS OF EXPERTISE

- Incident Response
- Root-Cause Analysis
- AWS (EC2, S3, VPC, IAM)
- Linux & Windows Server
- HAProxy, Nginx & Load Balancing
- Kubernetes (K3s) & ArgoCD GitOps
- Ansible, Python & Bash Automation
- Prometheus, Grafana, Elasticsearch & Alertmanager
- HIPAA / PCI-DSS Secure Design
- SIEM & IPS/IDS
- Terraform & Infrastructure as Code
- CI/CD (GitHub Actions, Jenkins)
- OpenStack, KVM & VMware vSphere
- PostgreSQL, MySQL, MSSQL & Redis

Kubernetes, ArgoCD, and OpenStack: self-hosted lab. Terraform and GitHub Actions: personal production site. All other tools, including Jenkins and Ansible: used at Atlantic.Net.

PROFESSIONAL EXPERIENCE

ATLANTIC.NET, Orlando, FL (hybrid) 12/2022 – Present

Continuous employment; part-time schedule 03/2025 – 11/2025 during the concurrent GE Vernova contract below.

DATA CENTER MSP TECHNICAL SUPPORT LEAD **07/2025 – Present**

- **Incident Command:** Escalation lead for 24/7 incident response on 1,200+ multi-tenant servers; cut monthly outages from 10 to 3–4 (ticketing-system data) and MTTR 25% by standardizing 15 runbooks.
- **Security Response:** Lead response to active exploitation campaigns on customer servers, including PHP2Shell (CVE-2026-109618, CVE-2026-109619).
- **Secure Architecture:** HIPAA/PCI-DSS environments (OPNsense, FortiGate, CrowdSec, IPS/IDS, SIEM) and multi-domain HAProxy tiers over Apache/Nginx/IIS with MSSQL, MySQL and PostgreSQL backends.
- **Deployment Automation:** Jenkins and Ansible playbooks deploy KVM VMs with user setup, SSH hardening and nftables rulesets; lead time for new managed environments fell from 18.8 to 5.6 days.
- **Platform & Team:** Operate the KVM/virsh platform (hundreds of VMs) with YAML provisioning and capacity planning; mentor 4–9 technicians on troubleshooting and on-call readiness.

ESCALATIONS SPECIALIST 09/2023 – 07/2025

- **Escalation:** Final escalation point for high-impact web, database and hybrid-cloud incidents (Nginx/Apache, PHP, Python; PostgreSQL, MySQL, MariaDB) within strict SLAs.
- **Network Isolation:** Traced cluster-wide broadcast storms to an ebtables misconfiguration on hypervisor nodes and restored tenant traffic isolation, cutting customer-reported performance issues by 50% (ticketing-system data).
- **Migrations and Modernization:** Led 100+ website migrations (cPanel, MySQL replication, 2-hour cutovers) and the Ansible CentOS 7 → Ubuntu 22.04/AlmaLinux upgrade of 200+ servers (60% less manual work, no configuration drift).

NOC HELP DESK TIER 2 12/2022 – 09/2023

- **Operations:** Monitored a 25,000 sq ft data center and 7 remote hosting sites for 200+ clients; resolved about 400-500 tickets a month, including work with Tier-3 ISPs.
- **Security:** Deployed Wazuh SIEM and pfSense/OPNsense firewalls (upgraded 300+ customer firewalls); managed CARP and IPsec; remediated vulnerabilities across 70%+ of client environments under HIPAA/PCI requirements.

SOGETI, CONTRACT AT GE VERNOVA, Remote (Melbourne, FL) 03/2025 – 11/2025

SYSTEM ADMINISTRATOR, PATCH MANAGEMENT *(concurrent with part-time work at Atlantic.Net)*

- **Patch Automation:** Python, Bash and PowerShell patch-lifecycle automation for 50+ production VMs (Red Hat 8, Oracle Linux 7, Windows Server) supporting power-grid EMS/SCADA; maintenance windows −30%, compliance 95%+, zero unplanned downtime.
- **Virtualization & Compliance:** VMware patch-test lab, vSphere/ESXi capacity planning, Veeam recovery and CIS benchmark audits; fixes deployed to live air-gapped systems.

EARLIER EXPERIENCE

IT CONSULTANT *Vendrameto Advogados & Aspen Chocolates (concurrent with judicial career)* 02/2012 – 06/2022

- Designed network segmentation (VLANs, firewall) and least-privilege Active Directory/GPO access for the law firm and for Aspen Chocolates, a manufacturer.
- Built and supported Windows Server infrastructure (Active Directory, file and print servers) and trained staff on Office 365 and OneDrive.

JUDICIAL LAW CLERK MANAGER *State of São Paulo Judiciary* 08/2015 – 07/2022

- Led 4 clerks and 2 interns from 2019 to 2022 (about 720 rulings a month); 100+ research projects cut average lawsuit resolution time 41% (17 to 10 months). Ran a Center for Conflict Resolution for 18 months (90% settlement rate).

PRODUCTION & LAB PROJECTS

- **Production portfolio site on AWS (jmarcelocarvalho.com):** Terraform-imported infrastructure with zero downtime and S3 remote state; GitHub Actions deploys through OIDC to SSM, with a tested rollback. Dockerized Flask on EC2 behind Nginx with PostgreSQL, Let's Encrypt TLS and automated S3 backups.
- **Kubernetes SRE lab (github.com/jmarcelo81/kubernetes_lab):** 4-node K3s cluster managed with Ansible and deployed through ArgoCD GitOps, with Traefik ingress, cert-manager, Vault, Prometheus/Grafana/Alertmanager and PatchMon.
- **Private cloud lab:** 3-node OpenStack on Dell PowerEdge servers with Ceph RBD, Neutron over VxLAN, Heat, RGW-Swift and Wazuh monitoring.
- **Infrastructure as code (github.com/jmarcelo81/terraform-multi-cloud):** Terraform provisioning on both AWS and OpenStack.

CERTIFICATIONS

Cloud & IaC: AWS Certified Solutions Architect – Associate (Jun 2026 – Jun 2029) · HashiCorp Certified: Terraform Associate 004 (Apr 2026 – Apr 2028) · AWS Certified Cloud Practitioner (Feb 2026 – Feb 2029) · CompTIA Cloud+ ce (Dec 2025 – Dec 2028) · CompTIA Secure Cloud Professional, CSCP (Dec 2025 – Aug 2028) · AWS Academy Cloud Architecting (Aug 2025)

Security & Foundations: CompTIA Security+ ce (Aug 2023 – Aug 2029) · CompTIA A+ ce (Oct 2022 – Aug 2029) · Microsoft SC-900 (Feb 2024) · Fortinet NSE 3, FortiGate 7.6 Operator (Apr 2026)

In Progress: AWS Certified AI Practitioner

EDUCATION & LANGUAGES

B.Sc. Computer Science: Faculdade Anhanguera, São Paulo, Brazil | Online program, studying alongside full-time work (in progress, expected December 2028)

Bachelor of Laws (LL.B.): Faculdades Integradas de Itapetininga (FII-FKB), Brazil

Languages: Portuguese (native) · English (fluent) · Spanish (C1) · Italian (B2)